



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE: PHYSICS
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mrs Govender
DATE	18 July 2023
MARKS	165
DURATION	2,5 hours

%

LEARNER NAME: MEMO

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 24 pages. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	Total
Mark										
Moderated Mark										
Question Total	14	24	11	19	33	23	15	11	10	165

Comment:

QUESTION 1 GR**14 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1 D ✓✓

1.2 C ✓✓

1.3 C ✓✓

1.4 B ✓✓

1.5 C ✓✓

1.6 D ✓✓

1.7 A ✓✓

QUESTION 2 KINEMATICS KH**24 MARKS**2.1 Define the term *displacement*. (1)

The change in position.✓

2.2 Calculate the magnitude of the acceleration of the vehicle as it moves from **A** to **B**. (3)

$$v^2 = u^2 + 2as \quad \checkmark$$

$$0,5^2 = 0^2 + 2a(5) \quad \checkmark$$

$$a = 0,025 \text{ m.s}^{-2} \quad \checkmark$$

2.3 Calculate the total time it takes for the vehicle to travel from **A** to **C**. (4)

A-B

$$v = u + at$$

$$0,5 = 0 + (0,025)(\text{COE 2.2}) t \quad \checkmark$$

$$t = 20 \text{ s} \quad \checkmark$$

A-B

$$v = s/t$$

$$0,5 = 4/t$$

$$t = 8 \text{ s} \quad \checkmark$$

Total time = 20 + 8 = 28s✓

2.4 Calculate the average speed of the vehicle during the trip from **A** to **C**. (2)

$$S = D/t = 9/28 \quad \checkmark = 0,32 \text{ m.s}^{-1} \quad \checkmark$$

2.5

The vehicle travels at constant speed throughout its motion at junction **B**.Did the vehicle accelerate as it turned left at junction **B**? Explain your answer. (2)

Yes✓. the direction of the vehicle changed✓ even though the speed remained constant.

2.5.1 Calculate the total distance travelled by the vehicle from **A** to **F**. (3)

$$D = 6 + 8 + 2(3) \quad \checkmark \checkmark = 20 \text{ m} \quad \checkmark$$

2.5.2 Determine the resultant displacement of the vehicle during the trip from point **A** to **F** as shown in the diagram. Give the direction as a bearing. (3)

$$r^2 = x^2 + y^2 = 8^2 + 6^2 = 10 \text{ m} \quad \checkmark$$

$$\text{Direction} = 90^\circ - 36,87^\circ \quad \checkmark$$

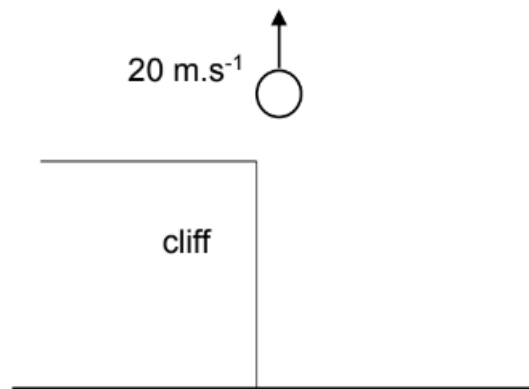
$$= 53,13^\circ \quad \checkmark$$

- 2.5.3 Calculate the magnitude of the average velocity of the vehicle during the trip from point **A** to **F**. (2)

$$V = S/t = 20/600 \checkmark (\text{COE 2.5.1}) = 0,03 \text{ m.s}^{-1} \checkmark$$

- 2.6 A golf ball travelling rectilinearly at a velocity of **Y** slows uniformly to come to rest after a distance **S**. At what initial velocity, **in terms of Y**, must it be struck to come to rest in **double the distance**, assuming that it experiences the same acceleration **a**? (4)

$v^2 = u^2 + 2as$	$v^2 = u^2 + 2as$
$0^2 = Y^2 + 2as_y \checkmark$	$0^2 = u^2 + 2a2s_y \checkmark$
$\frac{Y^2 = -2as_y}{u^2 = -2a2s_y} \checkmark$ OR making the accelerations equal to each other	
Solve in terms of u $u = \sqrt{2}Y \checkmark$	

QUESTION 3 KINEMATICS – KH**10 MARKS**

3.1 Define the term *velocity*. (2)

The rate of change of position. ✓✓

3.2 Calculate the time taken for the ball to pass the boy on its way downwards. (3)

$$v = u + at$$

$$-20 ✓ = 20 + (-9,8) t ✓$$

$$t = 4,08s ✓$$

3.3 Calculate the height of the ball, above the ground 2,8 s after it is released. (5)

$$s = ut + \frac{1}{2}at^2$$

$$= 20(6) + \frac{1}{2}(-9,8)6^2 ✓$$

$$= -56,4$$

$$= 56,4 m ✓$$

$$s = ut + \frac{1}{2}at^2$$

$$= 20(2,8) + \frac{1}{2}(-9,8)2,8^2 ✓$$

$$= 17,58 m \text{ above the cliff} ✓$$

$$\text{height above the ground}$$

$$= 56,4 + 17,58$$

$$= 73,98 m ✓$$

QUESTION 4 KINEMATICS FMC**23 MARKS**

4.1

4.1.1 Explain what is meant by the term free fall. (2)

(Motion in which) the only force✓ acting on the object is gravity (weight) ✓

4.1.2 Increasing velocity✓ down✓. (2)

4.1.3 Calculate the time that the golf ball is in contact with the floor before the first bounce. (1)

$$\Delta t = 0,67 - 0,64 = 0,03 \text{ s} \checkmark$$

4.1.4 Calculate the time it takes the golf ball to reach its maximum height after the first bounce. (2)

$$\Delta t = \frac{(1,90 - 0,67)}{2} \checkmark = 0,62 \text{ s} \checkmark$$

4.1.5 Calculate the speed at which the golf ball leaves the floor at the first bounce. (3)

Up+ (Accept any relevant equation and direction)

$$v = u + at$$

$$0 = u + (-9,8) \checkmark (0,62) \checkmark \text{ (COE 4.1.4)}$$

$$u = 6,08 \text{ m} \cdot \text{s}^{-1} \checkmark (6,076 \text{ m} \cdot \text{s}^{-1})$$

4.1.6 Calculate time **t** indicated on the graph. (5)

$$v^2 = u^2 + 2as \checkmark$$

$$0 = u^2 + 2(-9,8)1,2 \checkmark$$

$$u = -4 \text{ m} \cdot \text{s}^{-1} \checkmark$$

$$1,2 \checkmark = 4,85 t + \frac{1}{2}(-9,8)t^2 \checkmark$$

$$t = 2,95 \text{ OR } 2,96 \checkmark$$

OR $s = ut \frac{1}{2} + at^2$

$$1,2 = 0 + \frac{1}{2}(9,8)t^2$$

$$1,2 = 4,9t^2$$

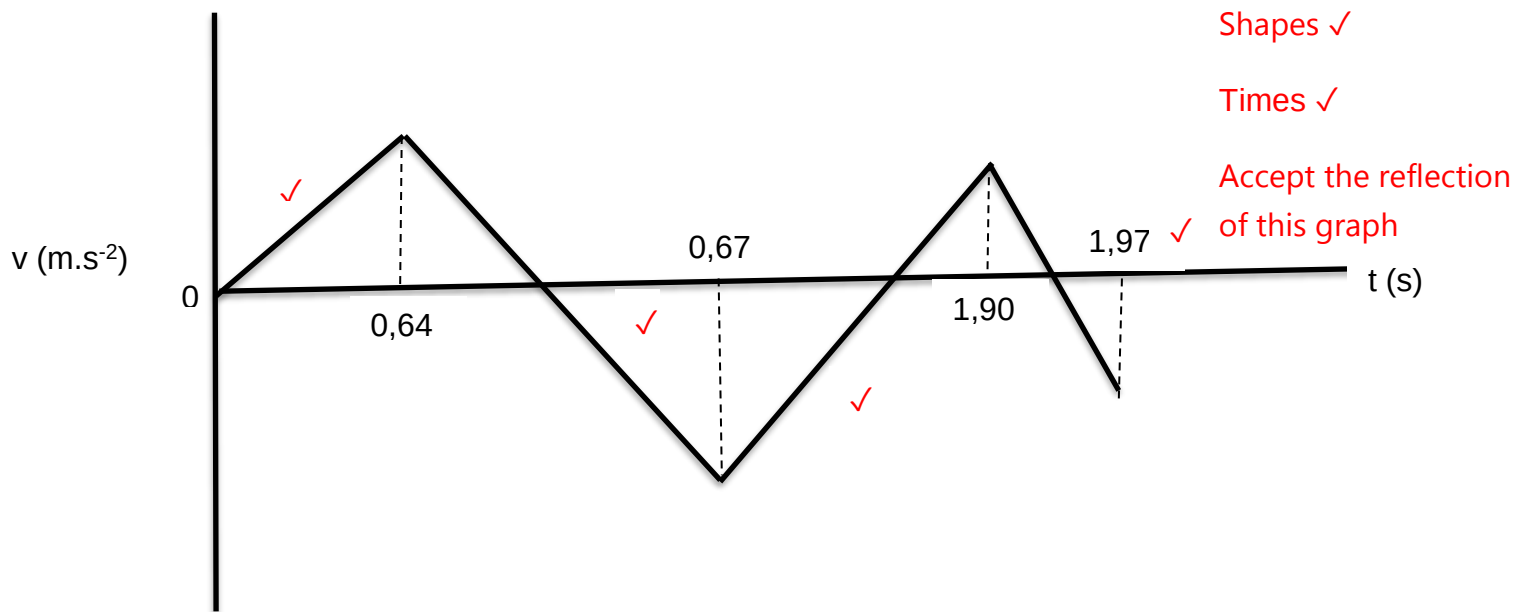
$$t^2 = 0,24$$

$$t = 0,49$$

$$Tt = 1,97 + 2(0,49)$$

$$t = 2,96$$

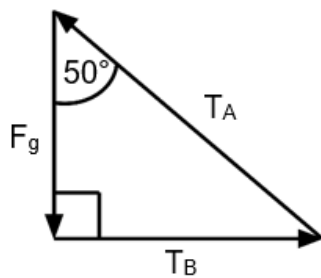
- 4.1.7 **Sketch** a velocity-time graph for the motion of the golf ball from time 0 s – 1,97 s. Clearly indicate the times of on the graph (no other values are required). (5)



- 4.1.8 Yes✓. The maximum height✓ reached by the ball decreases✓. (3)

QUESTION 5 NEWTON'S LAWS AND VECTORS FMC 9 MARKS

- 5.1 An object continues in a state of rest or uniform (moving with constant) velocity unless it is acted upon by a net or resultant force✓✓. (2)
- 5.2 (3)



One mark for each correct force (minus 1 for angle not shown)

- 5.3 Hence, determine the mass of the doll. (4)

$$\cos 50^\circ = \frac{F_g}{12,2}$$

$$mg = 12,2 \cos 50^\circ$$

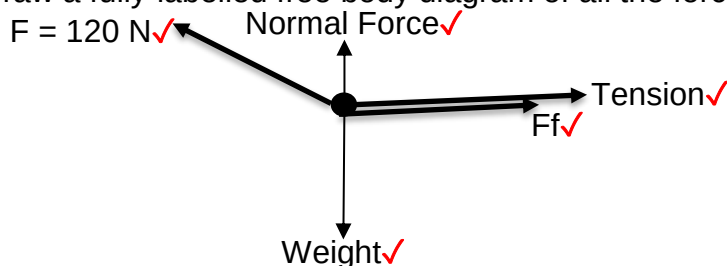
$$m = \frac{12,2 \cos 50^\circ}{9,8}$$

$$m = 0,8 \text{ kg}$$

QUESTION 6 NEWTON'S LAWS ZN**26 MARKS**6.1 Define the term *frictional force*. (2)

The force that opposes the motion of an object and acts parallel✓ to the surface with which the object is in contact ✓.

6.2 Draw a fully-labelled free body diagram of all the forces acting on the crate. (5)



6.3 Prove that the kinetic friction that the toy box is 2,76 N. (2)

$$F_{fk} = \mu_k F_N = 0,15 \times 167,87 \checkmark = 2,76 \text{ N} \checkmark$$

6.4 Calculate the normal force experienced by the cartoon and the crate if they have a combined mass of 25 kg. (3)

$$F_N = F_g - F_A = (25 \times 9,8) \checkmark - (120 \sin 40^\circ) \checkmark = 167,87 \text{ N} \checkmark$$

6.5 (3)

$$F_{fk} = \mu_k F_N \checkmark = 0,15 \times 167,87 \checkmark (\text{COE 6.4}) = 25,18 \text{ N} \checkmark$$

6.6 (6)

$$F_{\text{net}} = ma$$

$$F_A - F_f - T = ma$$

$$120 \cos 40^\circ - 28,18 - T \checkmark = 25a$$

$$T = 66,74 - 25a$$

$$T - F_f - F_{g\parallel} = 2a$$

$$T - (0,15 \times (2(9,8) \cos 20^\circ) \checkmark) - (2)(9,8) \sin 20^\circ \checkmark = 2a \checkmark$$

$$T - 2,76 - 6,7 = 2a \checkmark$$

$$66,74 - 25a = 2a + 9,46 \checkmark$$

$$57a = 27a$$

$$a = 2,12 \text{ m.s}^{-2} \checkmark$$

6.7 (3)

The net force would increase from 66,7 to 120✓. The friction would increase from 25 to 36,75✓. Yes, the net force would be greater ✓

QUESTION 7 NEWTON'S LAWS SG**19 MARKS**

7.1

- 7.1.1 Calculate the magnitude of the acceleration of the system and the magnitude of the tension in the cable. (5)

$$F_{net} = ma$$

$$FT - Fg = 5(a)$$

$$Fg - FT = 10(a)$$

$$FT = 5(a) + 5(9,8)$$

$$(10)(9,8) - T = 10(a)$$

$$FT = 5(a)✓ + 49 \dots (2)$$

$$(10)(9,8) - 10(a) = T$$

$$98 - 10(a)✓ = T \dots (1)$$

$$T - Fg5 = ma$$

$$98 - 10(a) = 5(a) + 49✓$$

$$T - (5)(9,8) = 5(3,27)$$

$$98 - 49 = 15(a)$$

$$T = 65,33 \text{ N}✓$$

$$a = 3,27 \text{ m.s}^{-2}✓$$

OR

$$Fg10 - T = ma$$

$$(10)(9,8) - T = (10)(3,27)$$

$$T = 65,33 \text{ N}✓$$

7.1.2 State *Newton's third law of motion*. (2)

When object A exerts a force on object B, object B **simultaneously** exerts an oppositely directed force of equal magnitude on object A. ✓✓

7.1.3 Give the name and direction reaction force to the weight of the 5 kg mass piece. (2)

The pulling gravitational force of the 5 kg mass piece on the Earth ✓ upwards ✓

7.2 Consider a brick on the inclined plane. The angle θ is increased gradually until the brick begins to slide.

7.2.1 Define the term *normal force*. (2)

The perpendicular force exerted by a surface on an object in contact with it ✓✓.

7.2.2 Write an expression in terms of θ for the normal force acting on the brick. (2)

$$F_N = mg \cos \theta \quad \checkmark \checkmark$$

7.2.3 Write an expression in terms of θ for the maximum static friction force ($F_{fs}^{\max}$) acting on the brick before it begins to slide. (2)

$$F_{fs}^{\max} = mg \sin \theta \quad \checkmark \checkmark$$

7.2.4 The 1,3 kg block is now placed onto a different rough surface with a coefficient of static friction of 0,48, inclined at an angle θ to the horizontal. The angle θ is now increased gradually, until the block starts to slide.
Calculate the angle θ when the block is just about to slide down the slope.
Show all your working out (4)

$$F_{fs}^{\max} = \mu_s F_N$$

$$F_{g\parallel} = \mu_s F_{g\perp} \quad \checkmark$$

$$mg \sin \theta = \mu_s mg \cos \theta \quad \checkmark$$

$$\mu_s = \frac{1,3 \times 9,8 \sin \theta}{1,3 \times 9,8 \cos \theta} \quad \checkmark$$

$$\tan \theta = 0,48$$

$$\theta = 25,64^\circ \quad \checkmark$$

QUESTION 8 ELECTRIC CIRCUITS SG**20 MARKS**8.1 State *Ohm's Law*. (2)

Current through a conductor is directly proportional to the potential difference ✓ across the conductor at constant temperature ✓.

8.2 Calculate the value of resistor **R** if the total resistance of the circuit is 4,8 Ω (3)

$$\frac{1}{4,8} = \frac{1}{4R} + \frac{1}{6R} \quad \checkmark \checkmark$$

$$Rs = 2\Omega \quad \checkmark$$

8.3 Define the term *potential difference*. (2)

The work done per unit positive charge. ✓✓

8.4 Calculate the reading on the voltmeter if the current through the 4R resistor is 1,8 A. (5)

$$I = \frac{V}{R} \quad \checkmark \quad V = IR = 1,2 \times 4 \quad \checkmark = 4,8 \text{ V} \quad \checkmark$$

$$I = \frac{14,4}{12} \quad \checkmark$$

$$I = 1,2 \text{ A} \quad \checkmark$$

8.5 Define the term *electrical work*. (2)

The work done on a charged particle by an electric field ✓✓.

8.6 Calculate the amount of energy dissipated by the **4R** resistor in 2 minutes. (3)

$$W = VIt = 14,4 \times 1,8 \quad \checkmark \times 120 \quad \checkmark = 3110,4 \text{ J} \quad \checkmark$$

The **4R** resistor is replaced with an ammeter.8.7 How will the reading on the voltmeter be influenced? Circle only INCREASE, **DECREASE** ✓ or STAY THE SAME. (1)

8.8 Explain the answer to QUESTION 7.7. (2)

The ammeter short circuits the resistors. ✓

No current flows through the resistor. ✓

QUESTION 9 ELECTRIC CIRCUITS ZN**11 MARKS**

The following circuit is set up using three light emitting diodes (LEDs); A, B and C to illuminate a small ornament. The amount of the light produced is determined by opening or closing switches S1 and S2. Resistors: R1, R2 and R3 all have an equal resistance.

9.1 Define *diode*. (2)

A component that will only allow current to flow in one direction ✓✓

The LED A has a knee voltage of 2,5V.

9.2 Briefly explain what a knee voltage is. (2)

The forward voltage required to allow a LED to work. ✓✓

9.3 Which switch should be closed to produce the brightest possible light?
Circle **S1** ✓ or S2. (1)

9.4 Explain your answer to QUESTION 8.3 (3)

R is equal so double the current will flow through R₁ compared to R₃ and R₂. ✓

Voltage is divided over B and C so A will receive double their voltage.

Power ≈ brightness and A receives 4 times the power or double the sum of B and C. ✓

9.5 If the emf of both cells are 9V what would the resistances of R3 and R2 need to be to ensure the potential differences over the LED B and C are equal to the knee voltage while allowing a current of 10 mA to flow through each LED? (5)

$$\text{Voltage remaining due to LEDs} = 9 - 2,5 - 2,5 \checkmark = 4V \checkmark$$

$$2R \checkmark = \frac{V}{I}$$

$$2R = \frac{4}{0,01 \checkmark} = 400$$

$$\boxed{R = 200 \Omega} \checkmark$$

QUESTION 10 ELECTROMAGNETIC SPECTRUM FMC 10 MARKS

10.1 How many unique frequencies will be in the emission spectrum as the excited electron returns to the ground state? (2)

3✓✓

10.2 Calculate the frequency of the photon emitted when the electron transition is from the $n = 2$ level to the ground state. (4)

$$E = hf \quad \checkmark$$

$$= (13,6 - 3,4) \quad \checkmark \quad 1,6 \times 10^{-19} = 6,6 \times 10^{-34} f \quad \checkmark$$

$$f = 2,47 \times 10^{15} \text{ Hz} \quad \checkmark$$

10.3 Hydrogen gas in a discharge tube emits light of wavelength 655 nm. Calculate the energy (in eV) that corresponds to this wavelength of light. (4)

$$E = \frac{hc}{\lambda} = \frac{(6,6 \times 10^{-34})(3 \times 10^8)}{(655 \times 10^{-9})} \quad \checkmark$$

$$E = 3,02 \times 10^{-19} \text{ J} \quad \checkmark$$

$$E = 1,89 \text{ eV} \quad \checkmark$$

ADDITIONAL SPACE

[illegible]
